EAGLE EYE

Airborne Intercept Radar Simulation

User Manual

Version 1.0.0 | April 2026

For use by virtual aviation squadrons operating with Microsoft Flight Simulator.

> ⚠ Eagle Eye is a radar simulation loosely based on the AN/APG-63 airborne intercept radar, using open-source and publicly available reference material only. It is not a certified or accurate simulation of any real system and must not be used for any purpose other than virtual aviation recreation.

1. Overview

Eagle Eye is a Windows companion application for Microsoft Flight Simulator (MSFS) that provides a radar simulation loosely based on an airborne intercept radar, presented on a standalone display window. It is designed for use by virtual aviation squadrons conducting simulated tactical intercept operations, providing radar situational awareness that is not natively available in MSFS.

The application connects to MSFS via the SimConnect SDK and renders a B-scope Multi-Function Display (MFD) with symbology, scan geometry, and detection physics loosely based on open-source analysis of the AN/APG-63 airborne intercept radar.

1.1 Key Features

- B-scope MFD with authentic phosphor green display and metal bezel
- Three radar modes: RWS (Range While Search), TWS (Track While Scan), STT (Single Target Track)
- Probabilistic detection model loosely based on open-source community analysis of the AN/APG-63
- Authentic scan geometry: 1, 2, 4, 6, and 8-bar patterns; ±30° and ±60° azimuth
- Ground Moving Target Rejection (GMTR) with selectable threshold
- Bullseye reference point with bearing/range readout
- OSB button interface matching the real MFD layout
- Automatic update checking on startup

1.2 System Requirements

Component	Requirement
Operating System	Windows 10 or Windows 11 (64-bit)
Microsoft Flight Simulator	MSFS 2020 or MSFS 2024 with SimConnect enabled
SimConnect DLL	SimConnect.dll and Microsoft.FlightSimulator.SimConnect.dll — included in application folder
.NET Runtime	.NET 8.0 or later (included with Windows 11; download from microsoft.com for Windows 10)
Display	Minimum 1280×720; Eagle Eye window is fixed at 560×620 pixels
Network	Required only for automatic update check — optional

2. Installation

2.1 Installing Eagle Eye

- Download the latest release ZIP from the Eagle Eye GitHub releases page.
- Extract the ZIP to a permanent location of your choice (e.g. C:\Program Files\EagleEye\).
- No installer is required. The application runs directly from the extracted folder.
- Ensure SimConnect.dll and Microsoft.FlightSimulator.SimConnect.dll are present in the same folder as EagleEye.App.exe. These are included in the release ZIP.

> ℹ Do not move EagleEye.App.exe out of the extracted folder. It must remain alongside the SimConnect DLLs.

2.2 First Run

- Launch EagleEye.App.exe.
- The application will attempt to connect to SimConnect automatically. Start MSFS before launching Eagle Eye if you want immediate connection.
- If MSFS is not running, Eagle Eye will retry the SimConnect connection silently in the background every 5 seconds until the simulator is detected.
- The SIM indicator on the right edge of the radar display will show boxed when SimConnect is connected.

2.3 Automatic Updates

Eagle Eye checks for newer releases on GitHub each time it starts. If a newer version is found, a prompt will appear offering to open the releases page in your browser. The check is silent if no update is available or if the network is unavailable.

2.4 Application Data

Eagle Eye stores two small settings files in your Windows AppData folder:

File	Contents
%AppData%\EagleEye\bullseye.json	Saved bullseye lat/lon coordinates
%AppData%\EagleEye\settings.json	Window preferences (Keep on Top)

3. Interface Overview

3.1 The MFD Window

The Eagle Eye window is a fixed 560×620 pixel display that cannot be resized. It replicates the general appearance of a Multi-Function Display with a dark metal bezel frame and five Option Select Buttons (OSBs) on each edge.

The display has four regions:

- Bezel — the dark outer frame containing the physical OSB buttons
- Scope glass — the active radar display area in the centre
- Left margin — elevation scale and left-edge OSB labels
- Right margin — range arrows, SIM indicator, and MENU label

3.2 OSB Button Layout

Buttons are numbered anti-clockwise from PB1 at the top-left of the left edge:

Buttons	Edge	Direction
PB1–PB5	Left edge	Top → bottom
PB6–PB10	Bottom edge	Left → right
PB11–PB15	Right edge	Bottom → top
PB16–PB20	Top edge	Right → left

3.3 Active Button Assignments

Button	Edge/Position	Label	Function
PB2	Left, 2nd from top	EL / N	Cycles bar pattern: 1 → 2 → 4 → 6 → 8 bars
PB4	Left, 4th from top	GMTR	Cycles GMTR filter: LO → MED → HI → CHAF → LO
PB7	Bottom, 2nd from left	RWS/TWS/STT	In RWS/TWS: toggles mode. In STT: breaks lock and returns to RWS
PB11	Right, bottom	MENU	Opens/closes the MENU page
PB14	Right, 2nd from top	▼	Range scale down
PB15	Right, top	▲	Range scale up

ℹ PB3 (left edge, 3rd from top) is labelled FS/2 on the scope. Frame Store selectability is planned for a future update.

4. Connecting to Microsoft Flight Simulator

Eagle Eye connects to MSFS automatically via SimConnect when the application starts. No manual action is required.

4.1 Connection Status

The SIM label on the right edge of the radar scope (aligned with PB13) shows the current connection state:

- SIM with a hollow green box — SimConnect is connected and MSFS data is being received
- SIM without a box — not connected (MSFS not running, or SimConnect unavailable)

4.2 Reconnection

If MSFS closes or SimConnect is interrupted, Eagle Eye will automatically attempt to reconnect every 5 seconds. The SIM box will disappear while disconnected and reappear when the connection is restored. No user action is needed.

4.3 Contact Sources

Eagle Eye reads two types of contact from MSFS:

- Own aircraft — your aircraft's position, heading, speed, and altitude are read at approximately 20 Hz via SimConnect's PLANE data variables.
- AI and multiplayer traffic — other aircraft visible to SimConnect within 148 km (80 nm) of your position are polled every 500 ms via RequestDataOnSimObjectType.

> ⚠ Eagle Eye reads the SimConnect TCAS contact table. Contacts are only available if MSFS is presenting them via that interface. Some third-party aircraft and multiplayer solutions may have varying levels of TCAS compatibility.

5. Radar Modes

Eagle Eye operates in three radar modes, selectable via PB7 or keyboard shortcuts. The current mode is shown in the bottom-centre of the display above PB7.

5.1 RWS — Range While Search

RWS is the primary search mode. The sweep bar moves across the scan volume at 120°/sec, illuminating contacts as it crosses their azimuth position.

How contacts appear

- A contact is only painted when the sweep bar crosses it — contacts do not appear continuously.
- First paint always shows as a brick (rectangle). The closure rate is not yet known.
- After two or more paints, the symbol changes: a downward-pointing triangle (hot triangle) indicates the contact is closing at more than 300 knots; an upward-pointing triangle (cold triangle) indicates it is opening at more than 300 knots.
- Painted positions are frozen — the contact symbol does not move between sweeps. It remains at the last painted position until the sweep crosses it again.
- Frame Store keeps contacts visible for up to two additional frame periods after their last paint, fading in brightness with age.

Entering RWS

- RWS is the default mode on startup.
- Press PB7 or key M to toggle between RWS and TWS.
- Breaking lock from STT always returns to RWS.

5.2 TWS — Track While Scan

TWS maintains persistent track files on contacts while continuing to scan. Up to 10 track files can be held simultaneously.

Track lifecycle

- Tentative — contact seen once; shown with dim symbology.
- Confirmed — contact seen on two or more consecutive sweep passes; shown with full symbology and a heading vector.
- Coasting — sweep passed but contact was not updated; track is extrapolated forward from last known velocity and shown dimmed.
- Dropped — contact not updated for two consecutive sweep passes; track file removed.

SPT Designation

- Up to 3 tracks can be designated as System Priority Tracks (SPT 1, 2, 3).
- Ctrl+click a confirmed track to designate it as the next SPT, or to remove an existing SPT designation.
- SPT 1 (PDT) is shown with a six-line asterisk. SPT 2 and 3 (SDT) are shown with a boxed diamond.

Entering and leaving TWS

- Press PB7 or key M to toggle between RWS and TWS.
- Track files are cleared when leaving TWS.

5.3 STT — Single Target Track

STT designates a single contact for precision tracking. The antenna slaves to the locked target in elevation; scan geometry constraints no longer apply in the vertical plane.

Entering STT

- Left-click a contact on the scope, or move the acquisition cursor over it and press Enter.
- STT can be entered from either RWS or TWS.

STT display

- The locked target is shown as a six-point star with a heading vector.
- Altitude is shown above the star in nearest thousands of feet.
- The top row shows: ground speed, aspect angle (e.g. 045R), and target heading.
- Range rate is shown on the right edge (e.g. 250> for 250 knots closing).

Breaking lock / leaving STT

- Press PB7, right-click, or Backspace to break lock and return to RWS.
- Lock drops automatically if the target exceeds the range scale or leaves the ±60° azimuth volume.
- Lock drops automatically if the target disappears from the SimConnect contact list (e.g. due to dropout or range limit).

5.4 Mode Reference

Action	PB7	Keyboard	Mouse
RWS ↔ TWS	PB7	M	—
Designate STT	—	Enter (over contact)	Left-click on contact
Break lock / STT exit	PB7	Backspace	Right-click
TWS SPT designate	—	—	Ctrl+left-click on track

6. Scan Geometry

6.1 Range Scale

The range scale sets the maximum displayed range. Valid scales are 10, 20, 40, 80, and 160 nautical miles. The current range is shown in the top-right corner of the scope.

Action	Button	Keyboard
Range up	PB15 (▲)	R or scroll wheel up
Range down	PB14 (▼)	F or scroll wheel down

6.2 Azimuth Width

The scan volume covers either ±30° (narrow) or ±60° (wide) of own aircraft heading. Narrow scan concentrates radar energy on a smaller arc for improved detection; wide scan covers more airspace.

- Toggle with key Z.
- The current azimuth width is not separately annunciated — it is visible from the sweep bar travel extent.

6.3 Bar Pattern

The bar pattern determines how many elevation bars the radar scans per sweep. More bars provide greater vertical coverage at the cost of a slower sweep repeat rate.

Pattern	Bars	Notes
1 bar	Single elevation bar	Fastest repeat rate; narrow vertical coverage
2 bar	Two bars	—
4 bar	Four bars	Default on startup
6 bar	Six bars	—
8 bar	Eight bars	Widest vertical coverage; slowest repeat rate

- Cycle bar pattern with PB2 or key B.
- Current bar count is shown adjacent to the EL label on the left margin (e.g. EL / 4).

6.4 Antenna Elevation

The antenna elevation offsets the entire scan volume up or down from the horizontal. The elevation caret on the left edge shows the current setting.

- Tilt up: key Q (+2° per press)
- Tilt down: key A (−2° per press)
- Range: ±60° from horizontal

The altitude coverage brackets shown to the right of the acquisition cursor indicate the maximum and minimum altitudes the current scan volume can detect at the cursor range, accounting for bar pattern and antenna elevation.

6.5 Frame Store

Frame Store (FS) determines how many historic sweep frames are retained on the display in RWS mode. Currently fixed at FS=2, meaning up to two previous frames are shown at reduced brightness alongside the current frame. This provides trail-like persistence of contact positions between sweep passes.

- FS=0: current frame only
- FS=1: current + 1 previous frame
- FS=2: current + 2 previous frames (current setting)

> ℹ Selectable Frame Store (FS 0/1/2) is planned for a future update.

7. Ground Moving Target Rejection (GMTR)

GMTR suppresses contacts whose ground speed falls below a selectable threshold. This filters out slow-moving ground vehicles, airport vehicles, and other surface traffic that would otherwise clutter the display during low-altitude operations.

Mode	Threshold	Use
LO	40 knots	Filters only the slowest ground vehicles
MED	80 knots	Default. Filters most ground traffic while retaining slow-moving aircraft
HI	160 knots	Aggressive filtering; may suppress slow or turning contacts
CHAF	200 knots	Maximum filtering; use to declutter in high-traffic environments

- GMTR Off is not available — a filter is always active.
- Cycle mode with PB4 or key G.
- Current mode is shown on the left margin (G/M/T/R column with active mode letters to the right).

8. Symbology Reference

8.1 Contact Symbols

Symbol	Mode	Meaning
Brick (rectangle)	RWS	First paint, or closure rate unknown
Hot triangle (▼, pointing down)	RWS	Closing at >300 knots (≥2 paints)
Cold triangle (▲, pointing up)	RWS	Opening at >300 knots (≥2 paints)
Diamond (outline)	TWS	Tentative or coasting track
Diamond (filled)	TWS	Confirmed track
Diamond (filled) + tick	TWS	SDT 2 or 3 (Secondary Designate Track)
Six-point asterisk	TWS	PDT 1 (Priority Designate Track)
Six-point star	STT	Locked target

8.2 Other Symbols

Symbol	Meaning
Heading vector	Short line from contact symbol in the direction of travel
Trail dots	Historical positions from previous sweep paints; fade with age
Chevron (own ship, bottom centre)	Own aircraft reference point
Acq symbol (two vertical bars)	Acquisition cursor — follows the mouse. In STT: locked to target position
Circle + centre dot	Bullseye reference point
Sweep bar (faint vertical line)	Current sweep position

8.3 Contact Colours

Colour	HAFU Classification
Phosphor green	Unknown (default)
Blue	Friend or Assumed Friend
Red	Hostile or Suspect

ℹ HAFU classification from IFF/scenario data is planned for Phase 2. All contacts currently show as Unknown (green) regardless of identity.

9. Display Annunciators

All labels and data fields are drawn inside the scope glass as part of the SkiaSharp renderer. No separate overlay windows are used.

9.1 Top Row (near scope border)

Field	Description
Three-digit bearing (top-left)	Absolute bearing to the acquisition cursor position
Two-digit range (below bearing)	Range to acquisition cursor in nautical miles
Range scale (top-right)	Current range scale in nautical miles (e.g. 40)
GS / ASP / HDG (STT only)	Target ground speed (knots), aspect angle (e.g. 045R), and heading

9.2 Left Margin

Label	Description
EL / N	Elevation / current bar count (e.g. EL / 4)
FS / 2	Frame Store — currently fixed at 2
G M T R / MODE	GMTR filter — right column shows active mode (LO, MED, HI, CHAF)
Elevation caret	Small horizontal tick marking current antenna elevation on the vertical scale
○ circles	Elevation circles showing beam coverage band above and below the caret

9.3 Right Margin

Label	Description
▲ (near top)	Range up indicator — aligned with PB15
▼ (below ▲)	Range down indicator — aligned with PB14
SIM (middle)	SimConnect status — boxed when connected, plain when disconnected
MENU (near bottom)	Menu page indicator — aligned with PB11
NNN> (STT only)	Range rate in knots. Positive = closing, negative = opening

9.4 Bottom Strip

Field	Description
G NNNN	Own aircraft ground speed in knots
G NNNN BBB-RR	Ground speed + bullseye bearing and range to acquisition cursor (when bullseye is set)

Field	Description
T NNNN	Own aircraft true airspeed in knots
HI (boxed)	PRF mode indicator — always HPRF in this simulation
N HI (inside scope, bottom-left)	Current bar number and PRF — e.g. '3 HI' = bar 3, HPRF. TWS shows 'TWSH'
RWS / TWS / STT (above PB7)	Current radar mode

9.5 Acquisition Cursor Altitude Brackets

When the acquisition cursor is over the scope in RWS or TWS mode, two numbers are shown to the right of the cursor:

- Upper number — maximum altitude (nearest thousands of feet MSL) detectable at the cursor range given the current bar pattern and antenna elevation.
- Lower number — minimum altitude under the same conditions.

If the cursor is within range of a contact, the contact's altitude is shown to the left of the cursor in nearest thousands of feet.

10. Bullseye

The bullseye is a fixed geographic reference point used for tactical position reporting. When set, a circle-and-dot symbol appears on the scope at the bullseye position, and the G field at the bottom of the display shows the bearing and range from the bullseye to the acquisition cursor.

10.1 Setting the Bullseye

- Press PB11 to open the MENU page.
- Press PB6 (SET BULLS) to enter the bullseye entry sub-mode.
- The display switches to an entry screen showing LAT and LON input boxes and a dial pad.
- Click the LAT box to make it active (bright border), then tap digits on the dial pad to enter the latitude. Use the − key for southern latitudes. Use the . key for the decimal point.
- Click the LON box and repeat for the longitude.
- Press BKSP to delete the last character in the active field.
- Press PB6 (CFM) to confirm. Eagle Eye will validate the entry (lat must be within ±90°, lon within ±180°). If the entry is invalid an error message appears — correct the values and press CFM again.
- Press PB8 (CXL) at any time to cancel and return to the MENU page without changing the current bullseye.

10.2 Clearing the Bullseye

- Press PB11 to open the MENU page.
- Press PB6 (SET BULLS) to enter the bullseye entry sub-mode.
- Press PB7 (CLR BULLS) to clear the bullseye immediately and return to the MENU page.

10.3 Bullseye Persistence

The bullseye coordinates are saved automatically whenever they are set or cleared. They are restored the next time Eagle Eye is launched. A confirmation message appears in the MENU page status area.

10.4 Using the Bullseye

When the bullseye is set and the acquisition cursor is visible on the scope, the G field at the bottom of the display shows: G [groundspeed] [bearing]-[range], where bearing and range are measured from the bullseye to the acquisition cursor position. This allows tactical position calls in the format 'bullseye bearing/range'.

> ℹ The bullseye symbol is only drawn when it falls within the current range scale and azimuth scan volume. It is suppressed outside those limits.

11. MENU Page

The MENU page is opened and closed with PB11. When the MENU page is active, the radar display is replaced by the MENU screen. Mouse and keyboard radar inputs are suspended. The SIM status line is shown on the MENU page.

11.1 MENU Page Buttons

Button	Label	Action
PB6	SET BULLS	Enter bullseye entry sub-mode (see §10)
PB7	ON TOP	Toggle Keep Window on Top. Label is boxed when active
PB11	RDR	Close MENU page and return to radar display

11.2 Keep Window on Top

When ON TOP is active (boxed label), the Eagle Eye window remains visible above all other applications including MSFS. This is useful when running Eagle Eye alongside the simulator on a single monitor. The setting is saved automatically and restored on next launch.

12. Keyboard Reference

ℹ All keyboard shortcuts are active on the radar page only. Keys have no effect while the MENU page is open.

Key	Action
R	Range scale up
F	Range scale down
B	Cycle bar pattern (1 → 2 → 4 → 6 → 8 → 1)
Z	Toggle azimuth width (±30° ↔ ±60°)
M	Toggle radar mode (RWS ↔ TWS)
G	Cycle GMTR filter (LO → MED → HI → CHAF → LO)
Q	Antenna tilt up (+2° per press)
A	Antenna tilt down (−2° per press)
Enter	Designate target under acquisition cursor (from RWS/TWS)
Backspace	Break STT lock / cancel
Left-click	Designate STT target / Ctrl+left-click to designate TWS SPT
Right-click	Break STT lock
Scroll wheel	Range scale up/down

13. Detection Model

Eagle Eye models detection performance using parameters loosely based on open-source community analysis of the AN/APG-63/70 radar. Detection is probabilistic — the same contact at the same range may be detected on one sweep and missed on another.

13.1 Key Parameters

Parameter	Value
Detection model	Pd = 1 / (1 + (R / R50)^6)
50% Pd range (5 m², nose-on)	72 nm
Maximum instrumented range	160 nm
Waveform	3-stage FMR — all three stages must detect for a contact declaration
Default RCS (unknown contact)	5 m² (fighter-sized)
Nose-on RCS multiplier	1.0× (maximum)
Beam-on RCS multiplier	0.30×
Tail-on RCS multiplier	0.20× (minimum)

13.2 Aspect Angle

Detection range is significantly affected by the aspect angle at which the radar sees the target. Nose-on (head-on geometry, 180°) provides maximum detection range. Tail-on (chase geometry, 0°) provides the shortest detection range because HPRF radar relies on closure Doppler energy, which is minimal from behind a target. Beam aspect (90°) is intermediate.

13.3 Bar Coverage

A contact must fall within the elevation gate of the bar currently being scanned to be detectable. If the contact is above or below the current scan volume it will not be painted regardless of range or aspect.

14. Known Limitations

14.1 VATSIM

When using Eagle Eye with a VATSIM-connected instance of MSFS, contacts will not be visible beyond the limitations of your pilot client. For vPilot, this can be configured in Settings > Performance and will have an absolute maximum of 100 nm.

- On range scales beyond your pilot client's configured range, the outer portion of the scope will be unpopulated even in high-traffic airspace. This is expected behaviour.
- Ensure your pilot client range is set to at least match your intended operating range scale in Eagle Eye.

14.2 JoinFS

When using Eagle Eye with JoinFS as the contact source, the maximum visible range is determined by the Circle of Activity radius configured in JoinFS Settings. Contacts outside that radius are not transmitted to the local MSFS session.

- Ensure the JoinFS Circle of Activity is set to a value at least as large as your maximum operating range scale. For Eagle Eye's maximum of 160 nm, a Circle of Activity of 160 nm or greater is recommended.
- The Circle of Activity setting is in the JoinFS application settings panel and requires a reconnect to take effect.

14.3 SimConnect Contact Data

- SimConnect AI/multiplayer contacts frequently report HeadingDeg = 0 regardless of actual heading. Eagle Eye derives heading from successive position updates in this case — heading vectors may be absent or inaccurate for the first few seconds after a contact appears.
- Contact speed data from SimConnect reflects True Airspeed, which may differ from ground speed at high altitudes. GMTR filtering uses ground speed where available.
- IFF discrimination is not implemented in this release. All contacts display as Unknown (green) regardless of identity.

14.4 Detection Model

- The detection model uses a fixed default RCS of 5 m² for all contacts regardless of aircraft type. Per-type RCS simulation is planned for a future update.
- The model does not simulate terrain masking, jamming, notching, or chaff effects.
- Detection probabilities are derived from open-source community analysis and are not representative of the classified performance of any real system.